Financial Mathematics: Exercises lecture 9

Ex 1. Prove the standard normal exponential-shift identity: for $Z \sim \mathcal{N}(0,1)$,

$$\mathbb{E}\left[e^{aZ-\frac{1}{2}a^2}\mathbb{1}_{\{Z>b\}}\right] = N(a-b).$$

Solution:

$$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2} \Rightarrow \mathbb{E}\left[e^{aZ-\frac{1}{2}a^2}\mathbb{1}_{\{Z>b\}}\right] = \int_b^\infty e^{az-\frac{1}{2}a^2}\phi(z)\,dz$$

$$= \int_b^\infty \frac{1}{\sqrt{2\pi}} \exp\Big(\underbrace{az-\frac{1}{2}a^2-\frac{1}{2}z^2}_{=-\frac{1}{2}(z^2-2az+a^2)=-\frac{1}{2}(z-a)^2}\Big)\,dz$$

$$= \int_b^\infty \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(z-a)^2}\,dz$$

$$\begin{cases} u := z-a \\ du = dz \end{cases}$$

$$= \int_{b-a}^\infty e^{-\frac{1}{2}u^2}\,du = 1-N(b-a) = N(a-b).$$

Ex. 2. Prove the put-call parity for $t<T$.

Solution:

The put-call parity is given by $\left[C-P = S-Ke^{-r(T-t)}\right]$.

Consider the portfolio:
- long call option
- short put option

- Value of this portfolio now (at time t) is: $C-P$.
- When at T, $S_T > K$, the long call will give a cashflow of $S_T - K$ and the short put a CF of 0 } TOTAL $= S_T - K$
- When at T, $S_T < K$, the long call will give a CF of 0 and the short put a CF of $-(K-S_T)$ } TOTAL $= S_T - K$
- When at T, $S_T = K$, both the put and call will give a CF of $0 = S_T - K$.

→ In all 3 cases, the CF is $S_T - K$ at time T.

So, the present value (at time t) of that CF should be equal to $C-P$:

$$C-P = PV_t(S_T-K) = S_t - Ke^{-r(T-t)}$$